

# Comparing Machine Learning Algorithms for Predicting Financial Inclusion in the East African Community

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## Abstract

Financial inclusion is a policy priority across the East African Community (EAC), yet a large share of adults remain outside the formal banking system. This study uses a real survey dataset of 23,524 individuals from Kenya, Rwanda, Tanzania and Uganda to (i) predict whether an individual owns a bank account and (ii) discover latent financial-inclusion segments. After cleaning and exploratory analysis, four supervised algorithms - Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA),  $k$ -Nearest Neighbours (KNN) and a feed-forward Neural Network - were trained and compared using accuracy, precision, recall, F1 and ROC-AUC. Because only 14% of respondents own an account, the target is highly imbalanced and accuracy is uninformative; F1 and ROC-AUC are therefore the decisive metrics. The Neural Network achieved the highest ROC-AUC (0.873) while QDA achieved the best F1 (0.500) at baseline. To improve performance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. This raised minority-class recall substantially for every model (e.g. the Neural Network from 0.375 to 0.677) and improved F1 for most, confirming a genuine performance gain on the metric that matters for identifying the unbanked. Unsupervised  $k$ -means clustering ( $k = 4$ ) recovered an interpretable inclusion gradient - from an older, rural, no-formal-education farming segment with  $\sim 9\%$  account ownership to a Kenyan self-employed segment near 21% - without using the target. Education, employment type, mobile-phone access and country emerged consistently as the strongest correlates of inclusion.

## 1 Introduction

Access to formal financial services underpins savings, credit and resilience to economic shocks. Across the EAC, mobile money has expanded rapidly, yet traditional bank-account ownership remains low and unevenly distributed. Identifying *who* is excluded, and *which* characteristics predict exclusion, helps target interventions. This paper frames financial inclusion as a supervised classification problem - predicting bank-account ownership from demographic and socioeconomic attributes - and complements it with unsupervised clustering to reveal natural population segments. Four algorithms of increasing flexibility are compared, a class-imbalance remedy is applied, and the resulting change in performance is measured. The contribution is a transparent, reproducible pipeline on a real regional dataset, together with an honest comparison of classical and deep-learning models under strong class imbalance.

## 2 Methods

### 2.1 Dataset

The data come from the FinScope financial-inclusion surveys and are publicly available on Kaggle.<sup>1</sup> They contain 23,524 individual records from four EAC member states: Rwanda (8,735), Tanzania (6,606), Kenya (6,068) and Uganda (2,101). Each record has a binary target, `bank_account` (Yes/No), and twelve attributes including country, survey year, location type (rural/urban), mobile-phone access, household size, age, gender, relationship to household head, marital status, education level and job type.

### 2.2 Data cleaning

Inspection revealed several data-quality problems, each handled explicitly: impossible survey years (2029, 2039, 2056) outside the 2016-2018 fieldwork window; an invalid education code (“6”); “don’t know / refuse” responses carrying no information; and household sizes of zero. These were set to missing. Records with a missing target (36 rows) were dropped. Remaining numeric gaps were imputed with the median and categorical gaps with the mode. The non-predictive identifier column was removed. The cleaned dataset has 23,488 complete records with no missing values.

### 2.3 Exploratory data analysis

The target is highly imbalanced: 85.9% of respondents have no bank account versus 14.1% who do (Figure 1). Account ownership rises steeply with education (from 3.9% for no formal education to 51-57% for tertiary/vocational) and with formal employment (up to 77.5% for government employees), and is strongly associated with mobile-phone access (18.4% versus 1.7%) and country (Kenya 25.1%, well above the other three) (Figure 2).

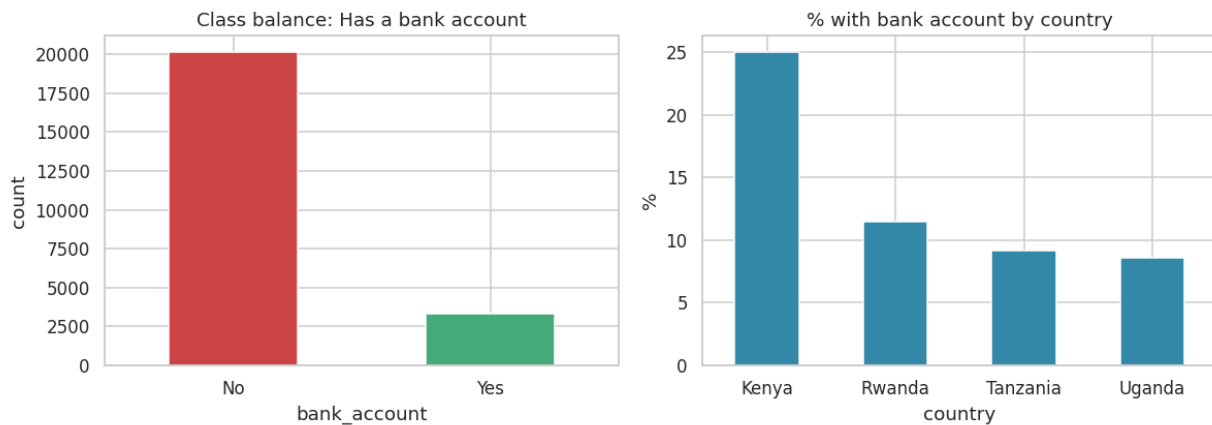


Figure 1: Class balance (left) and account ownership by country (right).

<sup>1</sup><https://www.kaggle.com/datasets/enockmokua/financial-dataset/data>

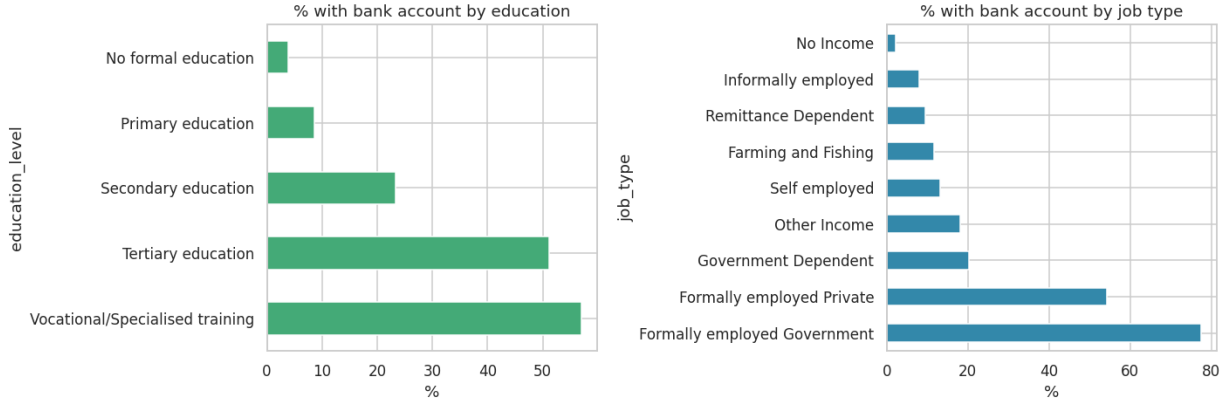


Figure 2: Account ownership rate by education level and job type.

## 2.4 Preprocessing and models

Categorical variables were one-hot encoded and the data split 80/20 with stratification. Features were standardised, since KNN, the discriminant models and the neural network are all scale-sensitive. Four supervised classifiers were trained to predict bank-account ownership, deliberately drawn from two distinct learning paradigms. The **classical machine-learning** models were: Linear Discriminant Analysis (LDA); Quadratic Discriminant Analysis (QDA, with covariance regularisation, `reg_param=0.5`, to avoid singular per-class covariance matrices arising from sparse dummy variables); and K-Nearest Neighbors (KNN,  $k = 15$ ). The **deep-learning** model was a feed-forward Neural Network implemented in TensorFlow/Keras, with two hidden layers (64 and 32 ReLU units), dropout (0.3, 0.2) and a sigmoid output, trained with the Adam optimiser and binary cross-entropy for 30 epochs. Using three classical algorithms together with one deep-learning model, this study satisfies the requirement to apply at least three different learning algorithms including deep learning; the Neural Network specifically provides the deep-learning component. All four models were evaluated on identical metrics—accuracy, precision, recall, F1 and ROC-AUC—so that classical and deep-learning approaches could be compared directly. Given the imbalance, F1 and ROC-AUC are treated as decisive. For reproducibility, a fixed random seed (47) was applied to all stochastic operations (the train/test split, SMOTE and  $k$ -means).

## 2.5 Improving performance

To address the imbalance, SMOTE was applied to the *training set only* (preventing leakage), synthesising minority examples until the classes were balanced. All four models were retrained on the balanced data and re-evaluated on the untouched test set.

## 2.6 Clustering

For the unsupervised task the target was removed and  $k$ -means applied to the standardised features. The number of clusters was selected using the elbow and silhouette methods, which agreed on  $k = 4$ .

### 3 Results and Interpretation

#### 3.1 Baseline classification

Table 1 reports test-set performance before any imbalance handling. All models reach high accuracy ( $\sim 0.88$ ), but this merely reflects the majority class; recall for the minority (account-holders) is low. The Neural Network attains the best ROC-AUC (0.873) and QDA the best F1 (0.500).

Table 1: Baseline performance (before improvement).

| Model          | Accuracy | Precision | Recall | F1           | ROC-AUC      |
|----------------|----------|-----------|--------|--------------|--------------|
| LDA            | 0.885    | 0.649     | 0.394  | 0.491        | 0.856        |
| QDA            | 0.866    | 0.528     | 0.474  | <b>0.500</b> | 0.843        |
| KNN            | 0.887    | 0.689     | 0.364  | 0.476        | 0.842        |
| Neural Network | 0.890    | 0.705     | 0.375  | 0.489        | <b>0.873</b> |



Figure 3: F1 and ROC-AUC across the four baseline models.

#### 3.2 After improvement (SMOTE)

Table 2 shows performance after retraining on the SMOTE-balanced training set. Recall for the minority class rises sharply for every model - the Neural Network from 0.375 to 0.677, LDA from 0.394 to 0.737 - and F1 improves for most models, with the Neural Network reaching the best overall F1 (0.545). Precision and headline accuracy fall, as expected, because the models now commit to identifying the minority class rather than defaulting to “No”. For a financial-inclusion use case, where the objective is to *find* unbanked or bankable individuals, this recall-oriented trade-off is the desirable outcome.

Table 2: Performance after SMOTE (after improvement).

| Model          | Accuracy | Precision | Recall | F1           | ROC-AUC |
|----------------|----------|-----------|--------|--------------|---------|
| LDA            | 0.797    | 0.385     | 0.737  | 0.506        | 0.863   |
| QDA            | 0.845    | 0.462     | 0.607  | 0.524        | 0.843   |
| KNN            | 0.766    | 0.346     | 0.746  | 0.473        | 0.835   |
| Neural Network | 0.841    | 0.456     | 0.677  | <b>0.545</b> | 0.870   |

Table 3: Before vs after: recall and F1 for the minority class.

| Model          | Recall |       | F1     |       |
|----------------|--------|-------|--------|-------|
|                | Before | After | Before | After |
| LDA            | 0.394  | 0.737 | 0.491  | 0.506 |
| QDA            | 0.474  | 0.607 | 0.500  | 0.524 |
| KNN            | 0.364  | 0.746 | 0.476  | 0.473 |
| Neural Network | 0.375  | 0.677 | 0.489  | 0.545 |

### 3.3 Clustering

The elbow and silhouette diagnostics (Figure 4) support  $k = 4$ . Table 4 profiles the segments. They form a clear inclusion gradient: an older, rural, no-formal-education farming segment with the lowest ownership ( $\sim 9\%$ ); a Kenyan self-employed segment with the highest ( $\sim 21\%$ ); and two mixed primary-educated rural/urban segments in between. Crucially, the clustering never uses the target, yet reproduces the same drivers found in the supervised analysis - evidence that the inclusion structure is intrinsic to the demographic profile.

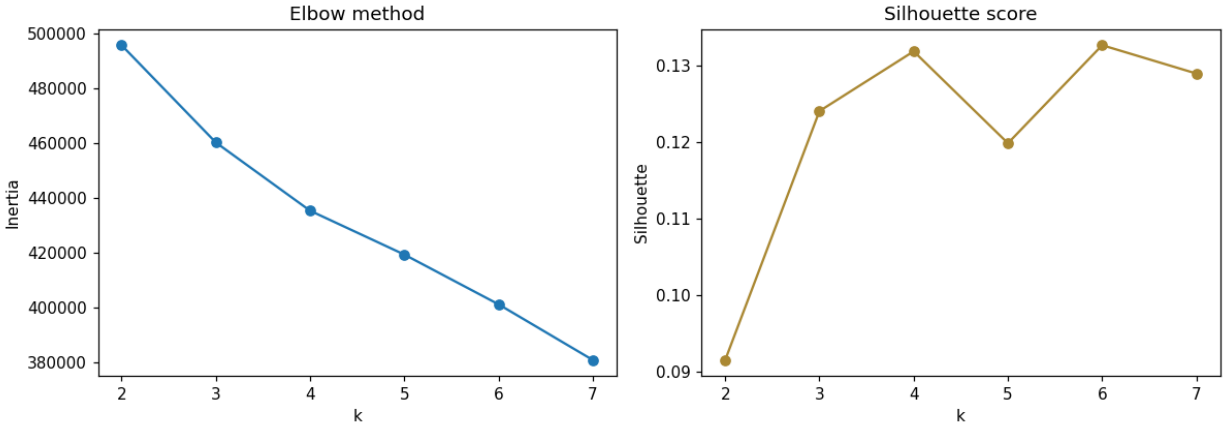


Figure 4: Elbow (left) and silhouette (right) diagnostics for choosing  $k$ .

Table 4: K-means segment profiles ( $k = 4$ ).

| Segment               | Size | Banked % | Mean age | Dominant country | Dominant job    |
|-----------------------|------|----------|----------|------------------|-----------------|
| Kenyan self-employed  | 5553 | 21.0     | 36.2     | Kenya            | Self employed   |
| Older rural farmers   | 2175 | 9.1      | 60.8     | Rwanda           | Farming/Fishing |
| Rwandan rural farmers | 5979 | 11.8     | 36.1     | Rwanda           | Farming/Fishing |
| Tanzanian urban       | 5083 | 11.3     | 35.7     | Tanzania         | Self employed   |

### 3.4 Worked example: single prediction

To illustrate the deployed model on an unseen case, one hypothetical respondent was passed through the identical preprocessing pipeline (one-hot encoding aligned to the training feature set, followed by standardisation) and scored by the deep-learning Neural Network. For a 30-year-old, urban, tertiary-educated woman in Kenya, formally employed in the private sector and with mobile-phone access, the model returned a predicted probability of 96.8% of owning a bank account, yielding a positive (“Yes”) decision at the 0.5 threshold. The prediction is consistent with the exploratory findings—urban location, tertiary education and formal employment are all strong positive correlates of inclusion—confirming that the model behaves sensibly on an individual, previously unseen profile.

## 4 Discussion & Recommendations

Three findings stand out. First, under strong class imbalance, headline accuracy is deceptive: every model exceeded 0.86 accuracy while missing most true account-holders. Reporting F1, recall and ROC-AUC is essential. Second, deep learning did not clearly dominate: the Neural Network led on ROC-AUC and, after rebalancing, on F1, but the far simpler discriminant models remained fully competitive on F1 and recall, consistent with the well-documented strength of classical methods on moderate-sized tabular data. Third, SMOTE delivered a real, interpretable improvement - large recall gains that make the models useful for locating the unbanked - at the cost of precision, a trade-off that should be tuned to the policy objective.

For practice, the consistent importance of education, formal employment and mobile-phone access suggests that financial-inclusion efforts should be coupled with education and, in particular, with mobile-based financial services, given the strong phone-access association. The excluded segment identified by clustering - older, rural, low-education farming households - is a concrete target for outreach.

Limitations include the vintage of the survey (2016-2018) and the exclusion of mobile-money ownership from the target definition, which may understate effective inclusion. Future work should incorporate the more recent FinScope 2024 microdata, add gradient-boosted trees as a strong tabular baseline, and tune the decision threshold explicitly against a cost model.

## 5 References

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